

CLASS 11-CBSE MATHEMATICS
CHAPTER: TRIGONOMETRIC FUNCTIONS-SET-1

TIME : 1Hr

Mx.Marks:30

SECTION A

Q.I Answer the following questions [1X3 = 3 MARKS]

1. If $\sin \theta + \cos \theta = m$ and $\sec \theta + \operatorname{cosec} \theta = n$, then find $n(m+1)(m-1)$.
2. If $\sin x = -\frac{24}{25}$, then find the value of $\tan x$.
3. A wheel makes 360 revolutions per minute. Through how many radians does it turn in 1 second?

SECTION B

Q.II Answer the following questions [2X5 = 10 MARKS]

4. Simplify $\frac{\sin^2 A - \sin^2 B}{\sin A \cos A - \sin B \cos B}$.
5. If $2 \tan A = 3 \tan B$, then find $\frac{\sin 2B}{5 - \cos 2B}$.
6. Prove that $2 \frac{\pi}{6} + \operatorname{cosec}^2 \frac{7\pi}{6} \cdot \frac{\pi}{3} = \frac{3}{2}$.
7. Prove that $\cos \cos \left(\frac{\pi}{4} - x \right) \cos \cos \left(\frac{\pi}{4} - y \right) - \sin \sin \left(\frac{\pi}{4} - x \right) \sin \sin \left(\frac{\pi}{4} - y \right) = \sin \sin (x + y)$.
8. Prove that $(\operatorname{cosec} x - \sin x)(\sec x - \cos x)(\tan x + \cot x) = 1$.

SECTION C

Q.III Answer the following questions [3X3 = 9 MARKS]

9. Find $\cos^{12} x + 3 \cos^{10} x + 3 \cos^8 x + \cos^6 x - 2$.
10. Prove that $\sin \sin (n+1)x \cdot \sin \sin (n+2)x + \cos \cos (n+1)x \cdot \cos \cos (n+2)x = \cos \cos x$.



11. Prove that $2x - 6x = \sin \sin 4x \sin \sin 8x$.

SECTION D

Q.IV Answer the following questions [4X2 = 8 MARKS]

12. If $\sin x + \cos x = m$, then prove that $\sin^6 x + \cos^6 x = \frac{4-3(m^2-1)^2}{4}$, where $m^2 \leq 2$.

13. Prove that $\frac{\sin 11A \sin A + \sin 7A \sin 3A}{\cos 11A \sin A + \cos 7A \sin 3A} = \tan 8A$.

